

e.g., $S = dReturn/dRisk$. The corresponding level of risk aversion on the efficient trading frontier at the best execution strategy X is $\lambda = dCost/dRisk$. This is also equal to the slope of the tangent at the point of the intersection of the cost-adjusted frontier and the ETF overlay. Notice that the slopes of the two tangents are approximately equal. Therefore, the corresponding level of risk aversion to ensure consistency across investment and trading decisions can be determined from the Sharpe ratio of the trade, e.g., $\lambda \cong S$. While this may not be the exact value it does at least ensure a large amount of consistency between investment and trading decisions. And it provides the trader with an appropriate input into the trade schedule optimization process which is extremely useful at times when they do not have enough time to perform a detailed analysis.

$$\begin{aligned} S &= dReturn/dRisk \cong dCost/dRisk = \lambda \\ S &\cong \lambda \end{aligned} \quad (10.7)$$

BEST EXECUTION FRONTIER

The next step in the portfolio construction process is for managers to consider the possibility that there may be a suboptimal Markowitzian portfolio, but after adjusting for trading costs and trading risk, this portfolio may in fact be pareto efficient with improved risk-return characteristics over the set of optimal Markowitzian portfolios. For example, is it possible that a portfolio that does not lie on the theoretical Markowitz efficient investment frontier, but after accounting for variable market impact cost and timing risk, the resulting cost-adjusted risk-return profile lies above the cost-adjusted frontier?

This problem is illustrated in [Figure 10.8](#). First, consider the three efficient portfolios on the efficient investment frontier (A_1, A_2 , and A_3). After accounting for trading costs, we arrive at the corresponding portfolios (x_1, x_2 , and x_3). The set of all cost-adjusted optimal portfolios results in the cost-adjusted frontier (described above). Next, consider the possibility that there exist a set of suboptimal portfolios (B_1, B_2 , and B_3). These are portfolios that do not initially lie on the efficient investment frontier but due to more favorable trading statistics (e.g., higher liquidity, lower impact sensitivity, lower volatility, etc.) they result in cost-adjusted portfolios (y_1, y_2 , and y_3) with higher risk-return characteristics. The resulting cost-adjusted portfolios correspond to a higher level of investor utility.

This frontier is defined as the best execution frontier. If these portfolios do in fact exist, investors would greatly benefit by investing in portfolios